Homework 1

MATH 441: DIFFERENTIAL EQUATIONS

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(Exercises are taken from *Elementary Differential Equations and Boundary Value Problems* by William E. Boyce and Richard C. DiPrima)

1. Exercise §2.1 #17. find the solution of the given initial value problem.

$$y' - 2y = e^{2t}, \quad y(0) = 2$$

$e^{\int -2\,dt} = e^{-2t+C_1} = C_2 e^{-2t}$

$\mu = e^{-2t}$

C_2 negligible as it is covered later in constant C

$e^{-2t}y' - 2e^{-2t}y = 1$

$e^{-2t}y = t + C$

$2 = C$

$y(t) = te^{2t} + 2e^{2t}$

2. Exercise §2.1 #19. find the solution of the given initial value problem.

$$t^3y' + 4t^2y = e^{-t}, \quad y(-1) = 0, \quad t < 0$$

$e^{\int 4t^{-1}} = Ct^4 \quad \mu = t^4$

$y' + 4t^{-1}y = t^{-3}e^{-t}$

$t^4y' + 4t^3y = te^{-t}$

$t \quad e^{-t}$

$1 \quad -e^{-t}$

e^{-t}

$t^4y = -te^{-t} - e^{-t} + C$

$0 = C$

$y(t) = -t^{-3}e^{-t} - t^{-4}e^{-t}$

3. Exercise §2.1 #40. Use the method of variation of parameters to solve the given differential equation

$$y' + (1/t)y = 3\cos 2t, \quad t > 0$$

$\int \frac{dy}{y} = \int -\frac{dt}{t}$

$\ln|y| = -\ln|t| + C_1 \qquad C_2 = e^{C_1}$

$y_g(t) = \frac{C_2}{t}$ var of para: $\frac{u(t)}{t} = y(t)$

$\frac{u'(t)t - u(t)}{t^2} = y'$

$\frac{u'(t)t - u(t)}{t^2} + \frac{u(t)}{t^2} = \cos 2t$

$u'(t) = 3t\cos 2t$

$u(t) = \frac{3}{2}t\sin 2t + \frac{3}{4}\cos 2t + C$

$y_g(t) = \frac{3}{2}\sin 2t + \frac{3}{4t}\cos 2t + \frac{C}{t}$

4. Exercise §2.2 #7. Solve the given differential equation

$$\frac{dy}{dx} = \frac{x - e^{-x}}{y + e^{y}}$$

separation of variables.

$$\int (y + e^{y})\, dy = \int (x - e^{-x})\, dx$$

$$\frac{1}{2}y^2 + e^{y} = \frac{1}{2}x^2 + e^{-x} + C$$

5. Exercise §2.2 #22. Solve the intial value problem

$$y' = 3x^2/(3y^2 - 4), \qquad y(1) = 0$$

and determine the interval in which the solution is valid.
Hint: To find the interval of definition, look for points where the integral curve has a vertical tangent.

$$(3y^2 - 4)\, dy = 3x^2\, dx \qquad 3y^2 - 4 = 0$$

$$y^3 - 4y = x^3 + C \qquad y = \pm\sqrt{\frac{4}{3}}$$

$$-1 = C$$

$$y^3 - 4y = x^3 - 1$$

$$\pm\sqrt{\frac{4}{3}}^3 \mp 4\sqrt{\frac{4}{3}} = x^3 - 1$$

$$\pm\frac{8\sqrt{3}}{9} \mp \frac{8\sqrt{3}}{3} + 1 = x^3$$

$$\sqrt[3]{\frac{(\pm 8 \mp 24)\sqrt{3} + 9}{9}} = x$$

$$\left(\sqrt[3]{\frac{(8 - 24)\sqrt{3} + 9}{9}}, \sqrt[3]{\frac{(24 - 8)\sqrt{3} + 9}{9}} \right) \approx (-1.2763, 1.5978)$$

since $x = 1$ lies between the points, the interval in between is selected.

6. Exercise §2.2 #33.

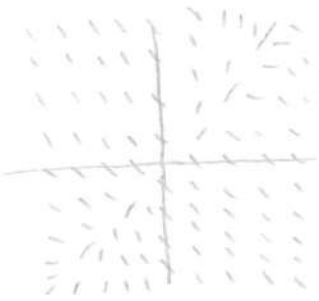
$$\frac{dy}{dx} = \frac{4y - 3x}{2x - y}$$

(a) Show that the given equation is homogeneous.

By definition of homogeneous equations, all terms in the equation above includes y, the unknown function, and therefore is homogeneous.

(b) Solve the differential equation

(c) Draw a direction field and some integral curves. Are they symmetric with respect to the origin?



drawn with desmos

7. Exercise §2.6 #10. Determine whether the equation is exact. If it is exact, find the solution.

$$(y/x + 6x) + (\ln x - 2)y' = 0, \qquad x > 0$$

exact, by theorem for condition of exact equations.